

The Village Electrician

Paper prototype report of a mouse-only game

1. Game Concept

Target Audience: Vocational education students studying electrical installations

Language: Based on professional vocational vocabulary (*Career Paths: Electricians* by Virginia Evans & Jenny Dooley).

Level 1 learning Objectives:

- Acquire authentic electrical terminology in English (*Subpanel, Neutral Bus Bar, Ground Bus Bar, Bonding Screw, Circuit Breaker*).
- Master practical safety codes: keeping neutral and ground isolated in a subpanel.
- Understand procedural tool application and safety-first workflows.

Control Scheme: Exclusively mouse-driven (Click to inspect, Drag-and-Drop to place, Click to use tools).

2. Narrative and Core Gameplay Loop

Narrative Setting: The player is an apprentice electrician arriving in an off-grid village where power is down. The previous village electrician retired, leaving behind an unfinished workshop, field notes, and scattered tools.

Mission Goal: Restore the electrical grid building by building, starting with the local barn subpanel, while strictly adhering to safety standards.

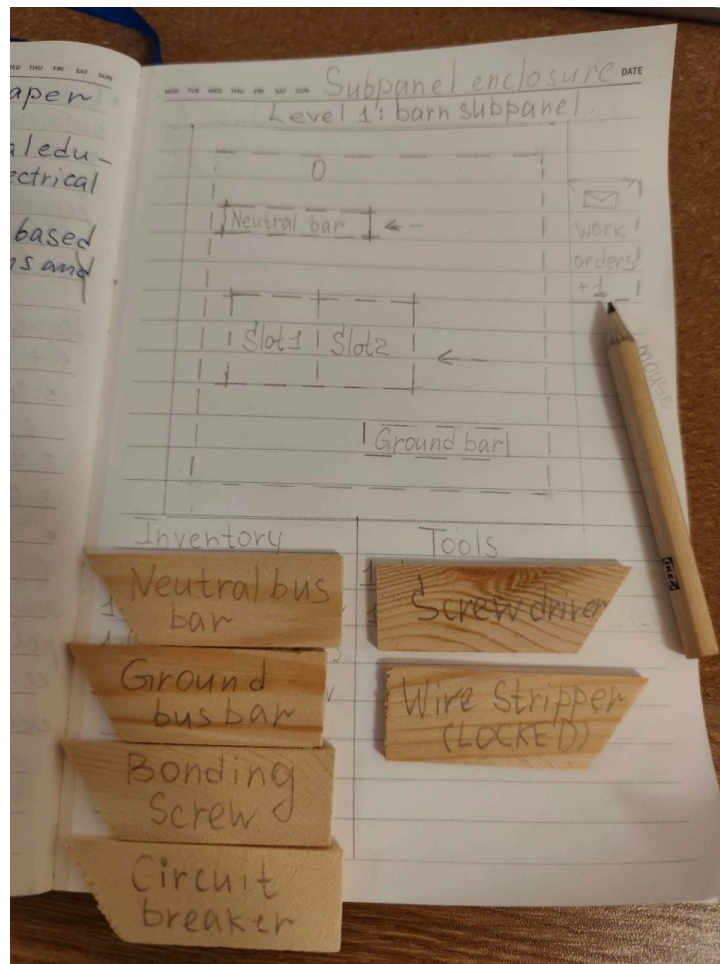
Core Gameplay Loop:

- **Accept Order:** Review client specifications via the work order system.
- **Explore and Learn:** Click environmental clues and diagnostic notes instead of reading lengthy text manuals.
- **Assemble:** Drag and drop components into the enclosure and secure them with tools.
- **Feedback:** Experience immediate hazard warnings upon errors and unlock new tools upon success.

Tool Progression: Completing early tasks unlocks specialized equipment (*Wire Stripper*), opening up new interactions for subsequent levels.

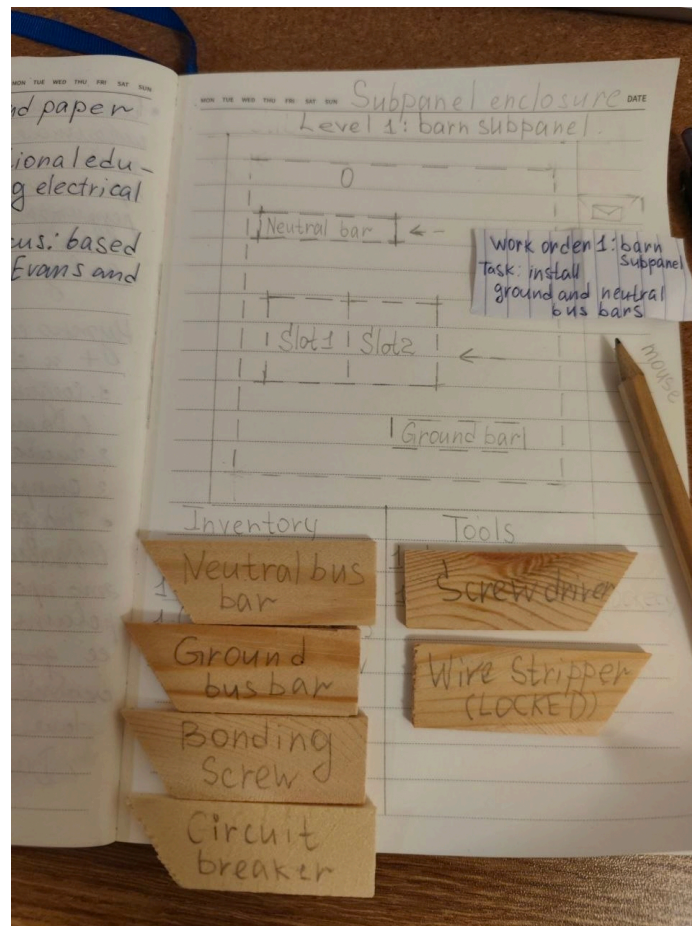
3. Action-by-Action Paper Prototype Session

Step 1: Workspace Overview



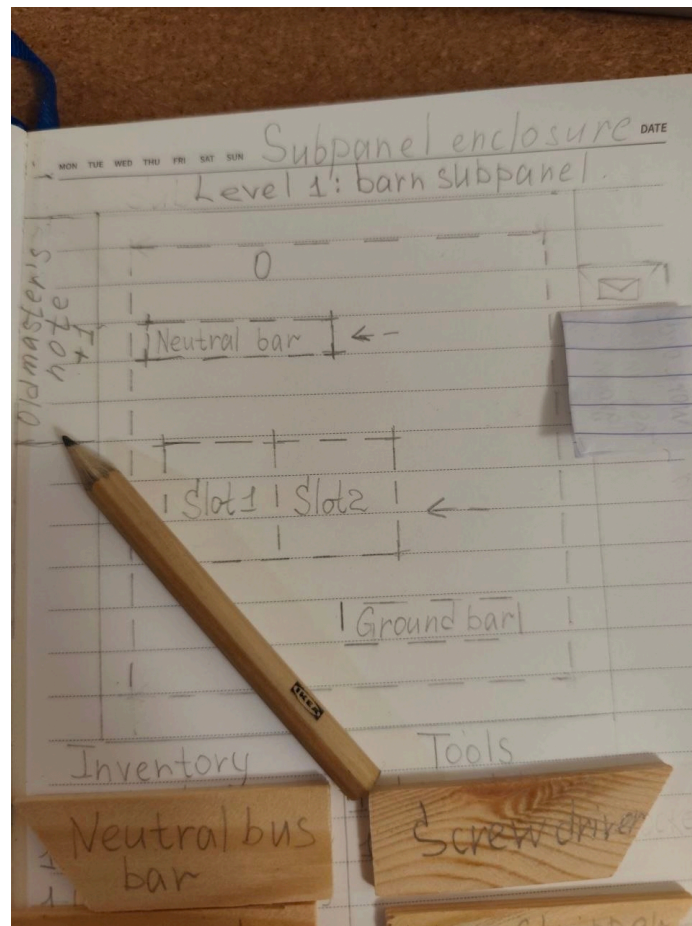
- The player examines the workspace and aims the mouse cursor at the incoming work orders envelope.
- The workspace displays the empty subpanel enclosure (Level 1: barn subpanel), component mounting areas, inventory dock, and tool shelf.

Step 2: Getting the work order



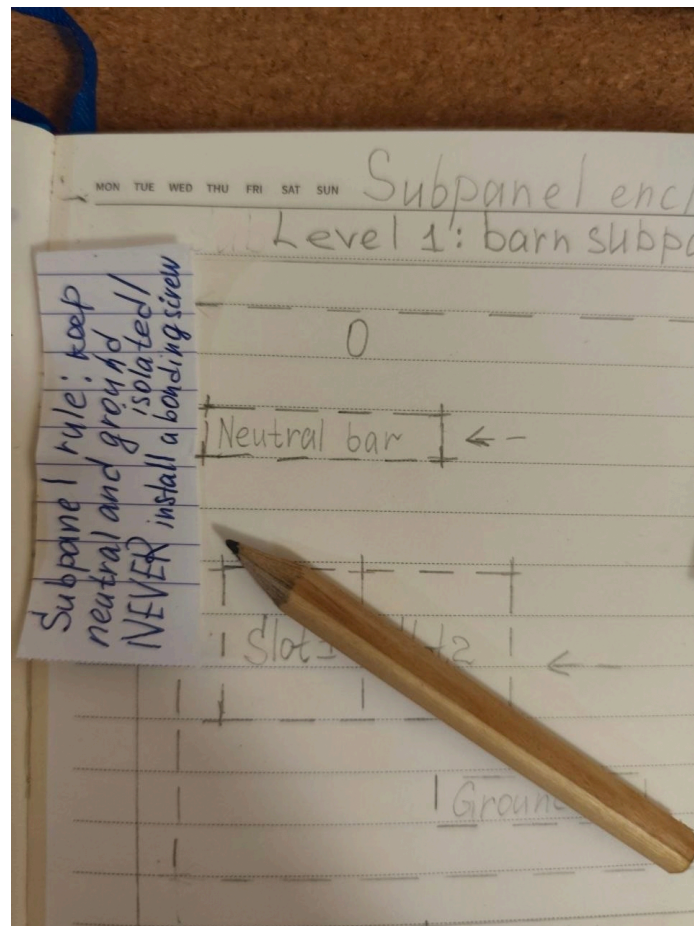
- The player clicks on the Work orders envelope using the mouse.
- The parts inventory is populated with required components (Neutral bus bar, Ground bus bar, Circuit breaker, Bonding screw), and the starting Screwdriver tool is made available. The Wire Stripper remains locked.

Step 3: Investigating clues



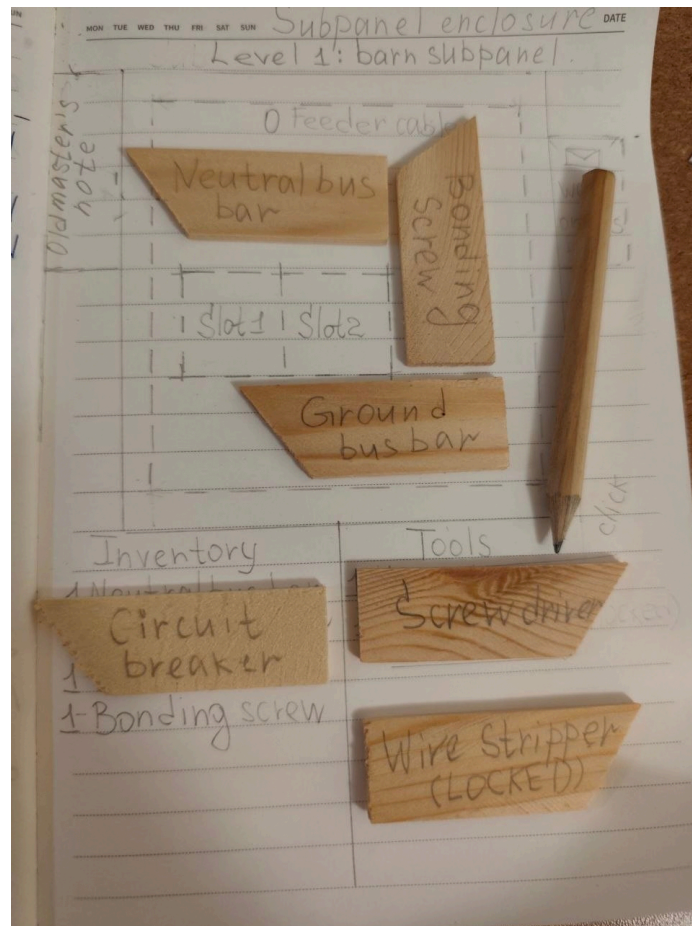
- The player clicks on the handwritten note pinned to the workshop wall (Old master's note).
- The system brings up contextual diagnostic documentation left by the retired electrician.

Step 4: Learning the Critical Safety Rule



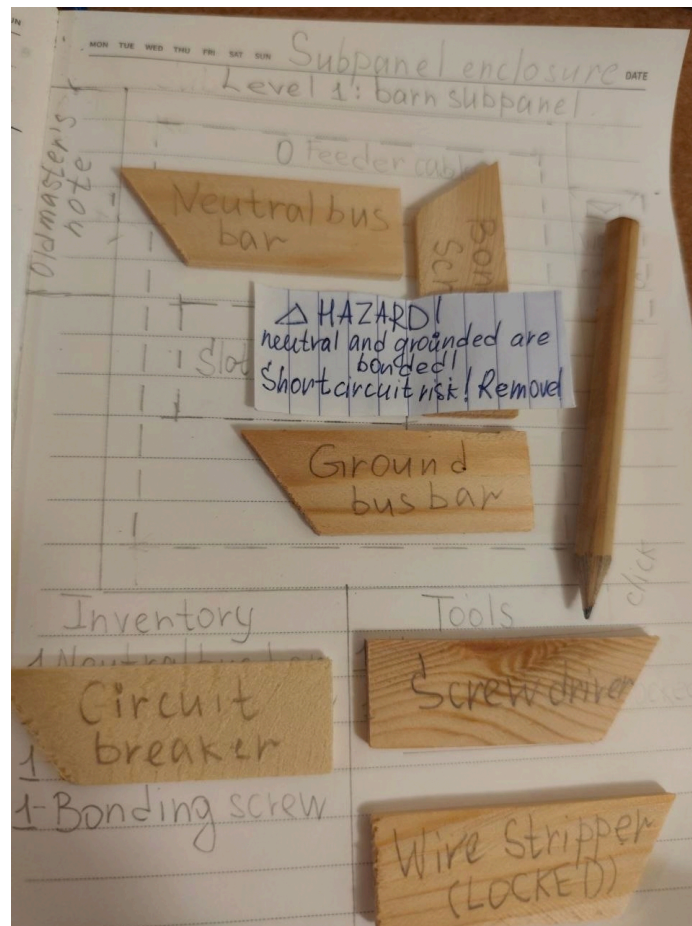
- The player reads the retrieved safety rule.
- The note reveals an essential safety principle: «Subpanel rule: keep neutral and ground isolated! NEVER install a bonding screw here».

Step 5: Component Assembly



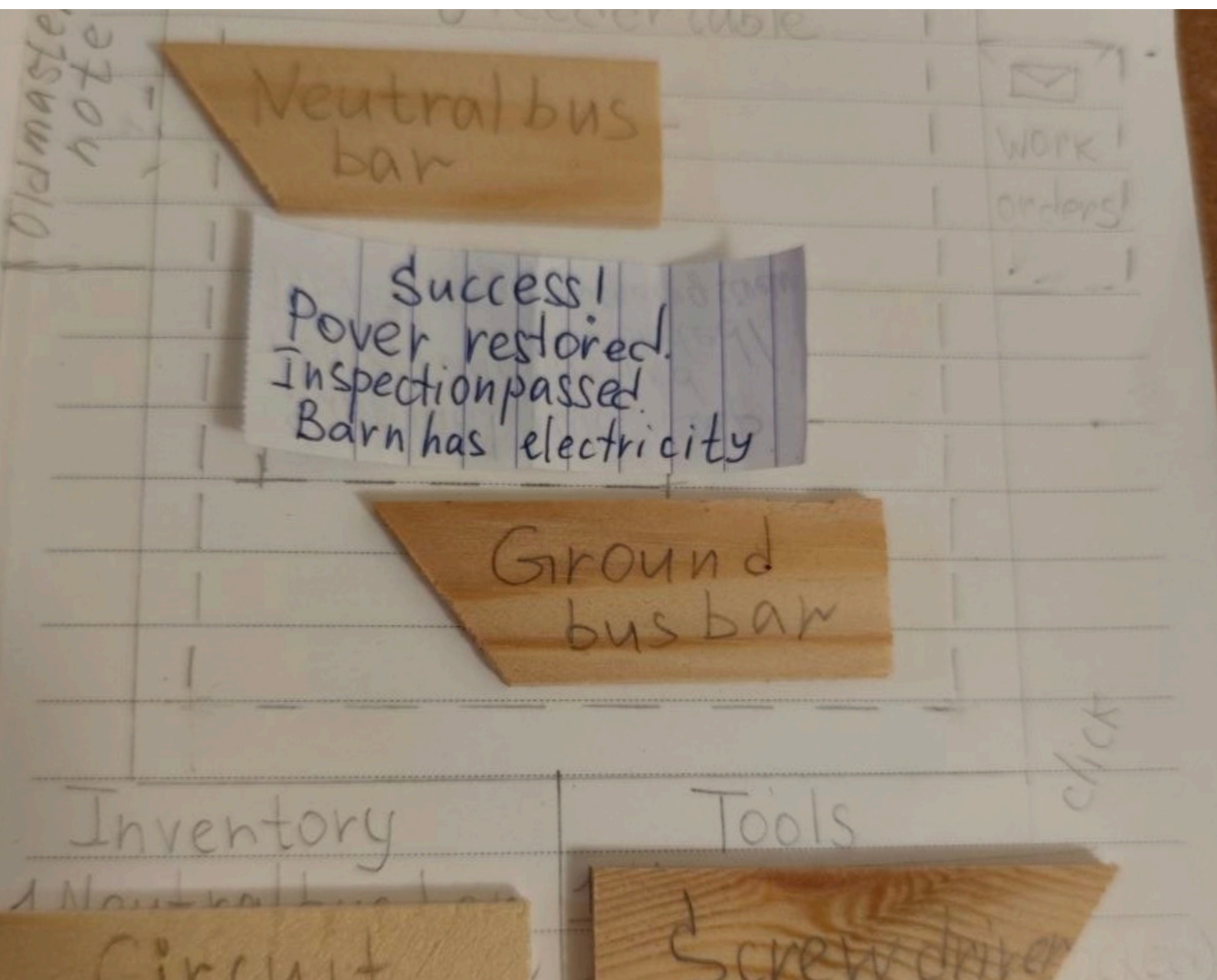
- The player drags the Neutral bus bar and Ground bus bar into the enclosure. However, the player also drags the Bonding screw into the neutral slot and clicks the Screwdriver to fasten it.
- The system detects a dangerous ground-neutral bond violating electrical codes (makes an unpleasant sound. Not sure about how to illustrate it).

Step 6: Hazard and Feedback



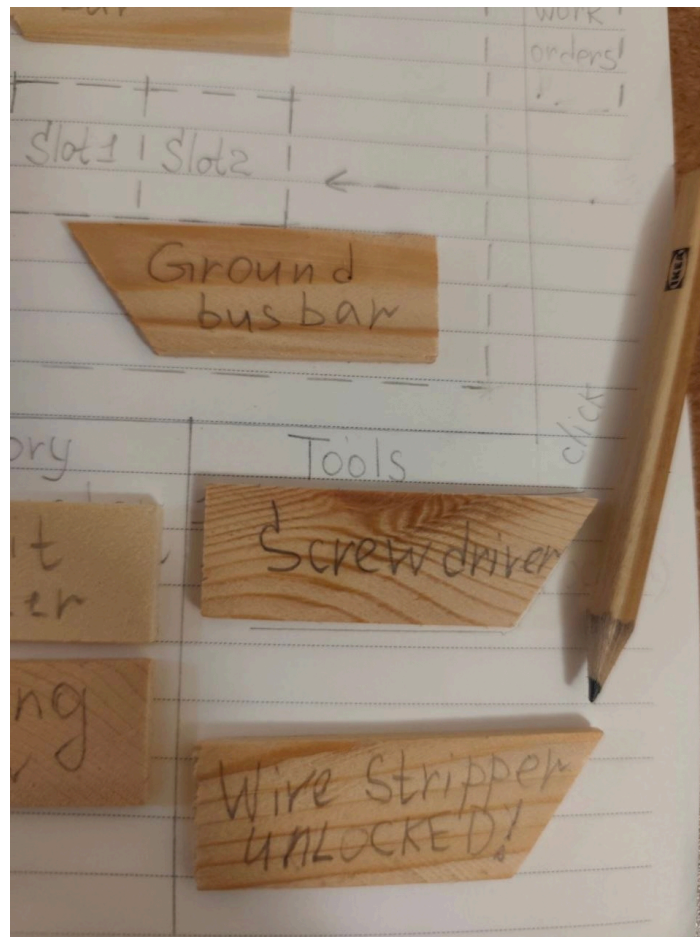
- The player encounters an immediate safety alert upon testing the installation.
- An urgent warning icon pops up: «HAZARD! Neutral and grounded are bonded! Short circuit risk! Remove!»

Step 7: Inspection Passed



- The player removes the bonding screw and completes the safe installation of the bus bars.
- The system validates the circuit and displays a success notification: «*Success! Power restored! Inspection passed! Barn has electricity*».

Step 8: Tool Progression and Level Complete



- The player clicks on the tools to claim the reward.
- The Wire Stripper switches to **UNLOCKED**, enabling wire-cutting and branch breaker mechanics for Level 2.